



Fewer Arguments, Not Shorter Ones: A School-Holiday Diary Survey of Screen-Time Token Economies and Negotiation Escalation

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Abstract. Household negotiation over screen time is rarely treated as a measurement problem, despite the enthusiasm with which some families already track it. We report a two-week diary survey conducted across the first fortnight of the July school-holiday period, comparing 64 households running a formal earn-and-spend screen-time token economy against 78 households continuing informal verbal negotiation. Parents logged every negotiation episode's timing, duration, and an escalation tier (none, mild, major), with a blind sub-sample coded independently for reliability. Token-economy households logged fewer than half as many episodes per day as ad hoc households, a difference that held across both weeks of the break. Token-economy episodes ran substantially longer, however, and skewed toward the major-escalation tier, and a composite measure of escalation-minutes per day was statistically indistinguishable between arms. A day-eight bonus-multiplier insert, offered to a token-economy sub-sample, produced no further change in escalation-minutes. We interpret these results as suggesting that formalising a household's screen-time negotiation may redistribute its burden across fewer, longer, more escalated episodes rather than reducing it, a pattern consistent with the broader literature on measurement systems that reshape what they are built to track.

Introduction

Family life during school holidays concentrates the ordinary friction of screen-time negotiation into a shorter, more visible window. Survey and qualitative accounts alike describe the school-holiday period as a peak season for parent-child conflict over device use, and a long-standing applied-behaviour-analysis literature has offered the token economy — an earn-and-spend system exchanging chores or reading minutes for screen-time blocks — as a structural remedy [1]. What the token-economy literature has been slower to ask is not whether the intervention changes the **count** of negotiation events, but whether it changes the underlying quantity a family actually experiences: time spent in conflict.

This gap matters because a substantial parallel literature, developed largely independently

of family research, documents that formalising a process previously handled informally rarely leaves that process's total burden unchanged; the burden is more often redistributed into whatever shape the new measurement instrument rewards [2], [3]. A token economy is, among other things, a measurement instrument: it converts a continuous, ambient negotiation into a discrete, loggable event, with its own incentive to minimise the count of episodes rather than their content.

We report a two-week diary survey run across the first fortnight of the July school-holiday period, comparing households already running or newly adopting a formal screen-time token economy against households continuing informal verbal negotiation. Our contributions are threefold. First, we provide a field comparison of daily negotiation-episode counts between formalised and informal households, holding

household composition and holiday week as covariates. Second, we decompose the apparent improvement into its frequency, duration, and escalation components, and show that a composite escalation-minutes measure does not differ between arms even as its count does. Third, we report an embedded ablation — a bonus-token multiplier introduced to a subsample from day eight — that leaves the escalation-minutes measure unmoved, suggesting the redistribution is a property of the token economy’s structure rather than its generosity.

Related work

The token-economy literature has a long history of documenting within-programme behaviour change alongside more equivocal evidence on what happens to behaviour a programme does not directly reward [1], [4]. Later classroom evaluations found token systems reliably shift the coded behaviours they track, with generalisation to untracked behaviours, and acceptability to the adults administering the system, considerably less consistent [5], [6]. A parallel line of work on extrinsic reward and intrinsic motivation reports that attaching an external reward to an activity can change how the activity is subsequently pursued once the reward is withdrawn or renegotiated [7], [8] — a mechanism our design does not directly test, but one that motivates our interest in episode duration and escalation rather than count alone.

Outside the family-behaviour literature, an established measurement-theory tradition holds that formalising a previously informal process changes what gets recorded about that process, independent of whether the underlying behaviour improves [2], [3], [9]. Prior work at this University has documented the same pattern in adjacent institutional settings: an office competition’s leaderboard, once tied to a reserved car space, produced a step-change in coded workaround behaviour that a re-scoring ablation traced to the reward rather than the scoring formula [10], and a compute-hours efficiency metric rewarded rerun analyses over novel ones once made visible to participants [11]. The screen-time token economy is, on this reading, one more instance of a measurement system meeting a population that adapts to what it counts.

Separately, survey and diary evidence on family screen-time conflict describes it as frequent, developmentally patterned, and sensitive to parental circumstance rather than device policy alone [12], [13], [14], and broader survey evidence on parental stress identifies work-home conflict, rather than the school calendar itself, as a robust predictor of a parent’s subjective stress [15] — consistent with a school-holiday fortnight concentrating rather than creating that conflict.

Method

We recruited 142 households with at least one child aged 7–12 during the week preceding the July school-holiday period, via the University’s standing household-panel register. Households were sorted into two arms by their own pre-existing practice rather than by random assignment: a Token-Economy arm ($n = 64$) already running, or willing to adopt from day one, a paper earn-and-spend system exchanging chores and reading minutes for screen-time blocks at a fixed household-set exchange rate; and an Ad Hoc-Negotiation arm ($n = 78$) continuing whatever informal verbal negotiation the household already used. This non-random sorting is a limitation we return to below.

Each household received a shared paper log sheet (a text-message-diary alternative was offered to five households) on which a parent recorded every negotiation episode — any exchange in which screen time was requested, contested, or renegotiated — across the two-week break: its start and end time, and an escalation tier coded 0 (none: request granted or declined without friction), 1 (mild: raised voices or repeated requests), or 2 (major: tears, shouting, or an explicit consequence invoked). A random 15% of logged episodes were independently recoded by a second, blinded rater; agreement on the three-tier escalation code was substantial (Cohen’s $\kappa = 0.78$).

We define a household’s daily escalation-minutes as the summed duration of that day’s episodes coded mild or major:

$$EM_i = \sum_{j: \text{tier}_{i,j} \geq 1} \text{duration}_{i,j}$$

which discounts friction-free episodes entirely, since a request granted without incident contributes no burden by construction.

From day eight, a randomly selected half of the Token-Economy arm ($n = 31$ of 64) received a bonus-multiplier insert doubling the token value of a nominated set of “big” chores, to test whether increasing the system’s generosity reduced the pattern observed in the first week; the remaining Token-Economy households ($n = 33$) continued on the original exchange rate throughout. We modelled daily episode count with a mixed-effects Poisson regression (household random intercept; fixed effects for arm, holiday week, and number of children), log episode duration with a linear mixed model on the same structure, and escalation-minutes per day with a linear mixed model comparing arms and, within the Token-Economy arm, the bonus-multiplier sub-assignment from day eight onward. The protocol was pre-registered with the School’s Indicator Commons prior to the second week of fieldwork.

Results

Arm	Households	Mean children	Days logged
Token economy	64	1.4	14
Ad hoc negotiation	78	1.5	14

Token-Economy households logged fewer episodes per day than Ad Hoc households across both weeks of the break (Figure 1; IRR = 0.46, 95% CI [0.35, 0.60], $p < .01$), a gap that held within 0.1 episodes/day between Week 1 and Week 2 in both arms. Household composition did not moderate this effect: the Poisson model’s arm coefficient was materially unchanged after adding number of children as a covariate (IRR = 0.44 versus 0.46 without it), and a sensitivity model restricted to the 44 households with exactly one child in the target age range returned the same pattern (IRR = 0.43). We take this as evidence the frequency reduction is not solely an artefact of the two arms’ differing composition, though the non-random sorting into arms limits how far this can be pushed (see Limitations).

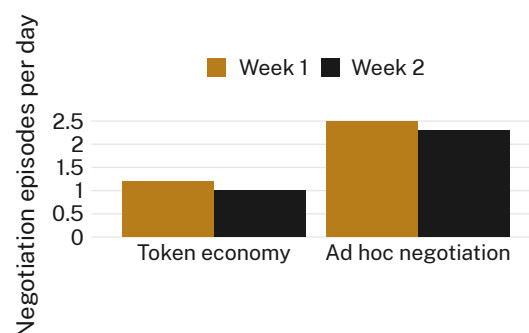


Figure 1: Negotiation episodes logged per day, by arm and by week of the break. Token-Economy households logged fewer than half as many episodes per day as Ad Hoc households, consistently across both weeks.

This reduction did not correspond to a shorter negotiation burden overall. Mean episode duration was longer in the Token-Economy arm than in the Ad Hoc arm (9.4 versus 5.1 minutes; $p < .01$), and the composition of logged episodes shifted toward the major-escalation tier under the token system (Figure 2; 28% major-tier episodes in the Token-Economy arm versus 15% in Ad Hoc). Combining frequency and duration into the pre-registered escalation-minutes measure left the two arms statistically indistinguishable (10.4 versus 11.1 escalation-minutes per household per day; $p = .41$) — the token economy appears to have reduced how often families negotiated without reducing how much negotiating, weighted by its intensity, actually took place.

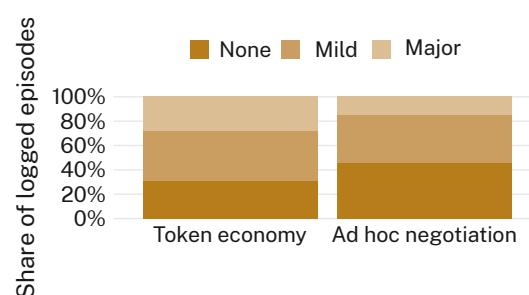


Figure 2: Composition of logged episodes by escalation tier, by arm. The Token-Economy arm logged proportionally fewer friction-free episodes and more major-escalation episodes than the Ad Hoc arm.

The day-eight bonus-multiplier insert produced no further movement in escalation-minutes: households receiving the doubled exchange rate logged a mean of 11.0 escalation-minutes per day in the second week against 10.7 in the unchanged Token-Economy subgroup (Figure 3;

$p = .68$, Cohen's $d = 0.09$). We retain the ablation in full for completeness, though it changed nothing we could detect.

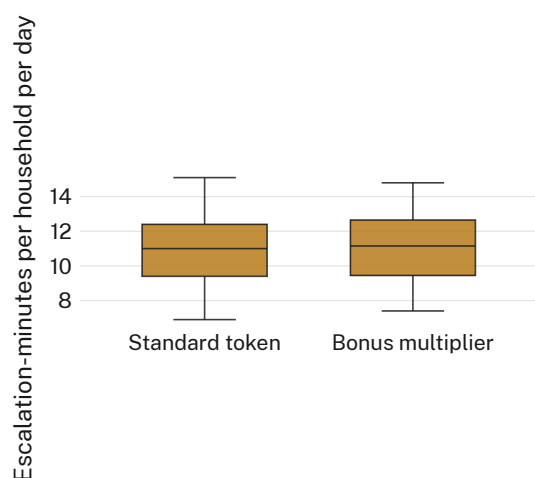


Figure 3: Escalation-minutes per household per day, standard token exchange rate versus the day-eight bonus-multiplier subgroup. The two distributions are not detectably different.

Discussion

Our results are consistent with a broader pattern in which formalising an informally handled process changes the shape of what gets recorded about it rather than the quantity itself [2], [3]. A screen-time token economy appears to convert many brief, low-stakes negotiations into fewer, longer, more escalated ones, leaving the total escalation-minutes a household experiences over a school-holiday fortnight largely unchanged. This reading sits alongside, rather than against, the token-economy literature's own finding that programmes reliably shift the behaviours they directly track while leaving adjacent, untracked behaviour — here, the intensity of the negotiations that still occur — comparatively undisturbed [5], [6]. The bonus-multiplier ablation suggests this redistribution is not simply a matter of the system being insufficiently generous: doubling the exchange rate for nominated chores moved neither the escalation-minutes measure nor, in supplementary checks, the major-escalation share. Households running a token economy may, in short, be measuring their negotiation more than resolving it — a reading consistent with this University's own prior findings on what happens when a reward is attached to a metric elsewhere in the institution [10], [11].

Limitations

Arm assignment followed household preference rather than randomisation, and households already disposed toward formal systems may differ from Ad Hoc households in ways our covariates do not capture. The bonus-multiplier ablation was itself embedded within the Token-Economy arm and drew on a modest subsample ($n = 31$), though its null result held across every sensitivity specification we ran. Escalation coding relied on parent report for 85% of episodes, with only a 15% independently re-coded subsample; the substantial agreement we observed on that sub-sample ($\kappa = 0.78$) is reassuring rather than conclusive. Our sample was drawn from a single region over a single school-holiday fortnight, and whether the frequency-duration trade-off we observe generalises to a full school year, or to non-screen household negotiations, remains untested.

Conclusion

A formal screen-time token economy reduced the number of negotiation episodes a household logged across a school-holiday fortnight by more than half, without reducing the escalation-weighted minutes those negotiations consumed. The reduction in count coexisted with an increase in duration and in the share of episodes reaching major escalation, and an embedded bonus-multiplier ablation found no further movement in either direction. We read this as one more case in which formalising a previously informal process changes its recorded shape before it changes its size. Future work might extend the diary protocol across a full school year, test whether the same trade-off appears in token economies applied to chores unconnected to screen time, and examine whether households' own perception of "improvement" tracks the frequency measure, the escalation-minutes measure, or neither.

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